

The Three Clocks of Live Music Generation: Sustained GPU-Free MRT2 Inference on iPhone

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Abstract

A live generative-audio system has to satisfy three clocks at once: the model must produce tokens before its autoregressive deadline, the renderer must deliver samples without interruption, and the generated trajectory must remain musically valid for as long as the system runs. Most inference benchmarks measure only the first clock. We present an end-to-end deployment of the 230-million-parameter `mrt2_small` live-music model on an iPhone 15 Pro Max. A single fixed-shape Core ML temporal graph executes all 12 transformer layers on the Apple Neural Engine (ANE), while the host owns a 41-frame K/V ring; a one-call 12-level RVQ rollout, SpectroStream decoder, inverse STFT, and lock-free audio ring complete the 48 kHz stereo pipeline. Ten of ten cold processes admitted the temporal graph to the ANE. A model-attributed Instruments trace recorded 653 temporal and 24 decoder ANE predictions and no application-attributed GPU interval.

The compute and delivery clocks pass under an unusually strict condition: a 610.19-second foreground run that began in the `serious` thermal state produced audio at 1.031 times real time, with zero underruns or drops and a 21.66 ms p99 effective cost against the 40 ms deadline. Per-minute p99 remained between 21.21 and 23.08 ms, and the audio reservoir grew rather than drained; a matched CPU/GPU-policy control reaches 49.23 ms p99 and drains its banked reservoir after minute six. The generative clock does not pass. Although the 600-second capture remains finite, stereo-coherent, prompt-aligned, and embedding-close to a clean reference, its last-window 4-16 Hz envelope energy is 0.136 against a frozen 0.070 limit, and only two of five blinded calibrated audio-model votes accept excerpts spanning the run. A temperature reduction removes the envelope excess but fails four other quality measures. We retain this as a result rather than tune the gate away.

The older A14 device exposes the hardware boundary: even with a 136 MB FLOAT32 decoder chosen to avoid an FP16 numerical failure, p99 is 58.59 ms, production is 0.897 times real time, and the maximum 20.16-second reservoir underflows after 294.08 seconds. Six post-training weight-compression variants shrink packages to 25-50% of baseline but all fail deterministic parity before device testing. These results show both what is solved and what is not: GPU-free, thermally sustained live-model inference on a phone is practical, but a throughput pass is not evidence of sustained generative quality. We release the exporters, fixtures, gate verifier, machine-readable receipts, and paper source.

1 Introduction

Live music generation is a systems problem with an audible failure mode. A batch image model can finish late; a live model that misses a deadline clicks, stutters, or falls silent. Magenta RealTime established the model class: a causal generator steered continuously by text or audio rather than a prompt that returns a fixed song [Lyria Team et al., 2025]. Magenta RealTime 2 (MRT2) extends that system with open weights and an Apple-Silicon implementation. Its small configuration contains 230 million parameters [Google, 2026] and emits 12 residual vector-quantized codes at 25 Hz; SpectroStream reconstructs 48 kHz stereo audio. Upstream targets Apple-Silicon Macs through MLX [Hannun et al., 2023] and a C++ runtime. This work asks a narrower, harder deployment question: what does it take to run the complete loop continuously on a phone without using the GPU?

The obvious success criterion is one frame every 40 ms. It is necessary and incomplete. A system can benchmark below 40 ms while silently executing on the wrong processor. It can average faster than real time while p99 stalls drain a finite buffer. It can deliver every sample on time while the autoregressive trajectory collapses into periodic clicking. We therefore define three clocks:

1. **Compute clock.** The temporal, depth, sampling, and decoder stages must produce each 25 Hz frame with p99 below 40 ms, or the result must be labeled as a buffering tier.

2. **Delivery clock.** Produced PCM must remain ahead of the audio callback for the full run, with zero underruns and drops and a non-draining reservoir.
3. **Generative clock.** The audio itself must remain finite, stereo-coherent, prompt-aligned, free of periodic defects, and perceptually acceptable across the complete horizon.

This separation is the paper’s central methodological claim. It prevented two false conclusions in one experiment. First, an early 600-second WAV appeared catastrophically broken even though live playback was correct; the evidence recorder had flattened a padded `MLMultiArray` instead of following its strides. Second, after that recorder was fixed, the runtime passed every throughput and delivery requirement, but blind excerpts and a frozen envelope metric found a genuine long-horizon audio defect. A single "real-time factor" would have hidden both failures.

We make four contributions:

- **A complete GPU-free phone deployment of MRT2’s compute path.** The temporal transformer’s full 12-layer math and SpectroStream decoder are observed on the ANE; depth rollout, sampling entropy, RVQ lookup, buffering, and DSP remain deliberate CPU work. We do not call this "all ANE."
- **A simple streaming boundary that is reliable in the application process.** K/V tensors are ordinary model inputs and one-token updates are ordinary outputs. Swift owns a preallocated 41-frame chronological ring. The graph is pure fixed-shape math and is admitted in 10/10 cold processes on both A17 Pro and A14.
- **A sustained evaluation that keeps the clocks separate.** On A17 Pro the compute and delivery clocks pass for ten foreground minutes from a hot start; the generative-quality clock fails its predeclared gate. On A14 a maximum reservoir cannot hide the thermally sustained deficit.
- **Negative results with early stopping.** Three compression methods across temporal and depth packages remain finite and much smaller, yet all fail deterministic state/token gates. FP16 decoder placements fail differently on the two phones. No rejected artifact is promoted by a speed-only result.

Our intended scope is a systems study, not a claim about a new generative model. The weights are unchanged. Previous conversion and ANE-admission findings are described in the companion *Surgical Inference* manuscript [Mireles, 2026]; this paper cites those methods and contributes the composed runtime, reliability study, sustain envelope, audio-delivery evaluation, and the three-clock result.

$$\mathcal{C}_{\text{compute}} \equiv Q_{0.99}(t_{\text{eff}}) < 40 \text{ ms}, \quad (1)$$

$$\mathcal{C}_{\text{delivery}} \equiv (U = 0) \wedge (D = 0) \wedge (B_{\text{end}} \geq B_{\text{start}}), \quad (2)$$

$$\mathcal{C}_{\text{gen}} \equiv \bigwedge_{k=1}^K m_k \in \mathcal{A}_k. \quad (3)$$

Here U and D are callback underruns and producer drops, B is queued PCM, and each m_k is a predeclared audio measure with acceptance set $\mathcal{A}_k$. A complete live-system pass is the conjunction of all three clocks; no score from one clock substitutes for another.

2 Background and related work

2.1 Live and controllable music models

MusicGen [Copet et al., 2023] generates text- or melody-conditioned music from interleaved neural codec tokens, but it is designed around bounded generation rather than a perpetual interaction deadline. RAVE [Caillon and Esling, 2021] demonstrated high-quality 48 kHz neural audio synthesis substantially faster than real time on a laptop CPU. Magenta RealTime [Lyria Team et al., 2025] introduced "live music models" whose output is causal and whose style can be steered during generation. MRT2 [Google Magenta, 2026] supplies the open model and Mac runtime used here. Our contribution is not a competing model; it is a physical-phone system and a sustained evaluation contract.

2.2 Neural audio codecs

SoundStream [Zeghidour et al., 2022] established end-to-end residual-vector-quantized neural audio and reported a streamable smartphone-CPU codec. EnCodec [Défossez et al., 2023] extended neural compression to high-fidelity 48 kHz stereo. SpectroStream [Li et al., 2025] moves full-band multichannel reconstruction into the time-frequency domain and combines delayed channel fusion with an RVQ representation. MRT2 generates the first 12 SpectroStream

codebooks at 25 frames/s. Our host sums the corresponding 256-dimensional codebook rows, invokes the fixed-shape decoder, and performs inverse STFT and overlap-add natively. Codec *decoding* in real time is therefore a necessary subsystem, not evidence that the autoregressive generator meets its deadline.

2.3 Mobile heterogeneous inference

MLPerf Mobile [Reddi et al., 2022] showed why on-device performance must be evaluated as a complete software-hardware stack rather than inferred from accelerator specifications. Apple has published ANE-oriented transformer layouts and fixed-shape deployment patterns through Core ML [Apple Machine Learning Research, 2022, Apple, 2026]. On Qualcomm devices, `llm.npu` [Xu et al., 2025] similarly reconstructs model boundaries and schedules work by hardware affinity. Our workload differs from mobile LLM prefill: the batch is one temporal frame, the deadline repeats forever, and the user hears tail latency directly. These constraints favor one fixed graph per repeated stage, host-owned recurrence, and direct measurement of p99, placement, thermal state, and buffer depth.

3 System design

3.1 Workload and deadline

`mrt2_small` contains a 12-layer temporal transformer and a 12-level autoregressive depth transformer. At frame t , the temporal stack consumes the previous frame’s mean token embedding, source conditioning, and local attention history. The depth stack produces one code from each of 12 codebooks, feeding each selected embedding into the next level. The selected embeddings are summed for SpectroStream. One generated frame represents 1,920 stereo samples, so the exact deadline is 40 ms at 48 kHz.

The deployed pipeline is summarized in fig. 1.

Temporal and decoder models are loaded with `.cpuAndNeuralEngine` on A17 Pro; depth uses `.cpuOnly`. Sampling-noise generation, tensor bookkeeping, and audio DSP also use the CPU. No runtime decision depends on an average latency.

3.2 Host-owned temporal state

The temporal artifact is a single-frame, pure tensor function. Its ordinary inputs are one `[1,1,1024]` temporal vector, one `[1,1,256]` source vector, a static attention bias, and 48 K/V arrays representing self- and cross-attention state across 12 layers. It returns the temporal output and 48 one-token K/V updates. The host preallocates all cache arrays and owns the 41-frame ring.

This boundary is intentionally dull. In-graph state mutation had compiled in a probe and then failed ANE admission in the application. Moving mutation to the host deletes that compiler variable without deleting attention math. The runtime writes each update into the ring, presents chronological cache inputs on the next frame, and advances a valid-history bias. There is one temporal model, not 41 position-specialized packages.

The state proof has two parts. First, the same input on a fresh and warmed runtime must diverge; otherwise state may be write-only. Second, 64 predictions must match a frozen reference, including 23 predictions after the 41-frame ring wraps. Core ML correlation is 0.9995 on both phones, finite ratio is 1.0, and maximum absolute errors are 0.8725 on A17 Pro and 0.7635 on A14.

3.3 One-call depth rollout

Calling a 75 MB depth model 12 times per temporal frame repeatedly streams its weights and cannot meet a phone deadline. The deployed depth graph unrolls the 12 dependent levels inside one prediction. The host supplies a `[12,1024]` Gumbel-noise tensor and inverse temperature; the graph applies the fixed top-40 mask, selects codes, gathers the next embedding, and returns both 12 codes and the mean temporal-feedback vector. Randomness remains seeded and host-owned, while the large weight stream occurs once per frame.

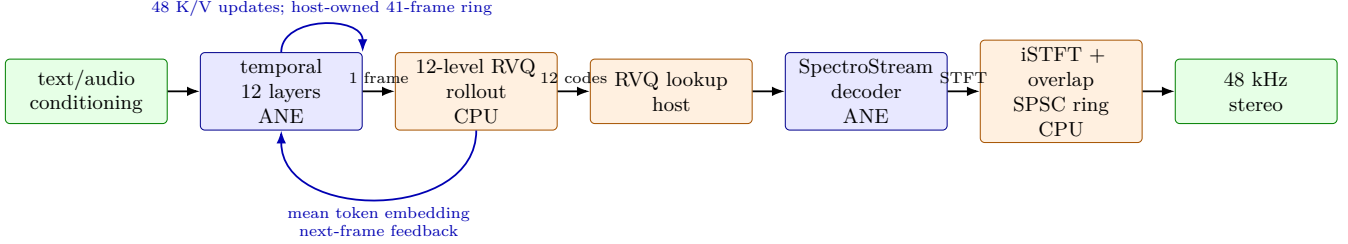


Figure 1: End-to-end deployment boundary. Blue stages are observed on the ANE; orange stages remain deliberate host/CPU work. Temporal recurrence crosses the model boundary as ordinary K/V tensors. No claim of an “all-ANE” system is made.

3.4 Decoder and real-time delivery

The first 12 RVQ codebooks are stored as a 12.6 MB float table. A host lookup sum forms each decoder embedding. The 68 MB FP16 decoder used on A17 Pro has channels-first internal geometry and a channels-last public STFT output. The output is materialized by following `MLMultiArray.shape` and `.strides`.

A producer thread invokes Core ML and pushes decoder STFT frames into a C++ render core. Inverse STFT and overlap-add occur off the audio callback. The callback performs only bounded ring reads and zero-fills an underflow; it never waits for inference. A finite startup reservoir absorbs ordinary jitter, but a native-real-time pass additionally requires the final reservoir not to be smaller than the initial one. This makes buffering visible rather than a way to hide a slow producer.

3.5 Bounded trajectory refresh

Short corrected captures were clean while unbounded trajectories deteriorated after roughly 90 seconds. We tested the smallest intervention: every 10 seconds the host clears only temporal K/V and previous temporal feedback. Decoder embeddings already queued, decoder overlap, and the audio ring remain intact. This separates temporal trajectory history from delivery continuity. At 180 seconds the intervention passes all objective bands, including 0.926 embedding similarity and 0.053 pulse share. At 600 seconds it does not prevent the late-horizon clicking failure. We therefore report it as a bounded ablation, not as a solution.

4 Experimental method

4.1 Devices and software

We evaluate a 2023 iPhone 15 Pro Max (iPhone16,2, A17 Pro, iOS 26.5.2) and a 2020 iPhone 12 Pro (iPhone13,3, A14, iOS 26.5). Runs use signed release builds in the foreground with the screen on. The prompt is **warm ambient texture**, source conditioning is a certified **warm.bin** vector, top-k is 40, style guidance is 3.1, and the principal capture uses temperature 1.0 and seed 20260718. The runtime logs every generation iteration, compute policy, thermal state, ring depth, pushed/pulled frames, underruns, drops, and stage time.

4.2 Effective frame metric

One generation-loop iteration produces 25 temporal/depth frames and one or occasionally two decoder batches. Backpressure sleep is not model work. We define effective frame time as

$$t_{\text{eff}} = \frac{t_{\text{temporal}} + t_{\text{depth}} + t_{\text{sampling}} + t_{\text{decoder}}}{25}.$$

We report run-level p50, p90, and p99. The 600-second gate requires at least 15,000 effective-frame samples. The dispersion campaign uses at least five independent application processes per device/policy cell and reports the median and IQR of run-level percentiles. Startup is measured from the auto-start event to the first audible PCM after load, placement/state proof, warmup, and priming.

4.3 Placement evidence

Requested compute units are metadata, not evidence. At session start, `MLComputePlan` must assign at least 0.95 estimated temporal cost to the ANE with zero GPU operations. Ten consecutive cold processes must preserve the same model hashes, policy, and state proof. A process-targeted Instruments trace must then contain temporal and decoder ANE intervals, Core ML intervals for temporal/depth/decoder, and no Metal GPU interval attributed to the application. This proves admission reliability and actual accelerator activity as separate properties.

4.4 Audio integrity

The 600-second float WAV is produced by an independent render core from the same logical STFT tensor used by playback. The gate checks 48 kHz stereo, finite ratio 1.0, clipping ratio at most $1e-5$, maximum normalized decoder chunk-boundary jump at most 0.07, left/right correlation at least 0.97, text embedding adherence at least 0.30, embedding similarity to a clean reference at least 0.80, and 4-16 Hz envelope-pulse power share at most 0.07.

Six frozen controls target known failure families: stride corruption, missing temporal feedback, write-only state, a click-comb capture, channel collapse, and injected dropouts. All must be rejected. Five 24-second lineups use unique frozen order seeds, neutral labels, RMS matching, and shared peak attenuation. Each contains the candidate, the clean MLX reference, and the known-bad click control. A multimodal audio model’s vote is valid only when it passes the known-good clip, rejects the known-bad clip, and ranks those controls correctly. The candidate requires at least four of five valid votes. These votes are a calibrated secondary instrument, not human-subjective ground truth.

4.5 Receipts and early stopping

Every public verdict is produced by a checked-in verifier from a machine-readable manifest. Private runtime logs and traces are hashed into those manifests. Precision experiments run in the order finite $\rightarrow$ deterministic parity $\rightarrow$ device latency $\rightarrow$ audio. A candidate that fails an earlier gate is not installed or listened to. This avoids both wasted measurement and the temptation to select a fast invalid artifact.

5 Results

Table 1: Sustained foreground results. The selected A17 policy passes compute and delivery; its CPU+GPU-policy control fails compute despite buffered delivery. A14 is a bounded-reservoir failure. Deadline: 40 ms.

Device	p50 effective	p90 frame	p99 (ms)	Rate ($\times$ RT)	First underflow	Verdict
A17 ANE	20.26	20.81	21.66	1.031	none / 610 s	pass
A17 CPU+GPU	23.56	43.55	49.23	1.008	none / 610 s	compute fail
A14 ANE	43.88	47.25	58.59	0.897	294.08 s	fail

5.1 Reliable ANE admission and GPU absence

Both devices pass 10/10 cold-process temporal plan/state gates. The A17 Pro trace contains 653 temporal ANE predictions totaling 5.007 s and 24 decoder ANE predictions totaling 0.341 s. It also contains 586 depth Core ML predictions, as expected for the CPU stage. The application’s Metal GPU table is empty: zero intervals and zero duration. The A14 supporting trace likewise contains temporal and decoder ANE intervals and an empty application GPU table for the FP16 artifact set used in that placement experiment.

The claim is therefore **GPU-free**, not **all-ANE**. The CPU remains essential for sampling-noise construction, the depth placement chosen for this runtime, RVQ lookup, state mutation, iSTFT, ring management, logging, and UI work.

5.2 Cross-process dispersion and the CPU/GPU-policy control

Table 2: Independent-process latency campaign, five runs per cell. Entries are median (IQR) across run-level percentiles. The CPU+GPU control changes only the requested temporal policy; depth remains CPU and decoder remains ANE.

Device	Temporal policy	p50	p99	Temporal p50	Startup
		ms/effective frame			s
A17 Pro	ANE	20.77 (.03)	22.35 (.42)	11.25 (.01)	4.26 (.04)
A17 Pro	CPU+GPU	27.32 (.25)	38.51 (1.78)	16.35 (.15)	6.22 (.05)
A14	ANE	40.38 (1.00)	50.72 (3.30)	13.85 (.63)	7.52 (.01)
A14	CPU+GPU	49.73 (1.11)	50.64 (1.41)	24.85 (.17)	9.76 (.14)

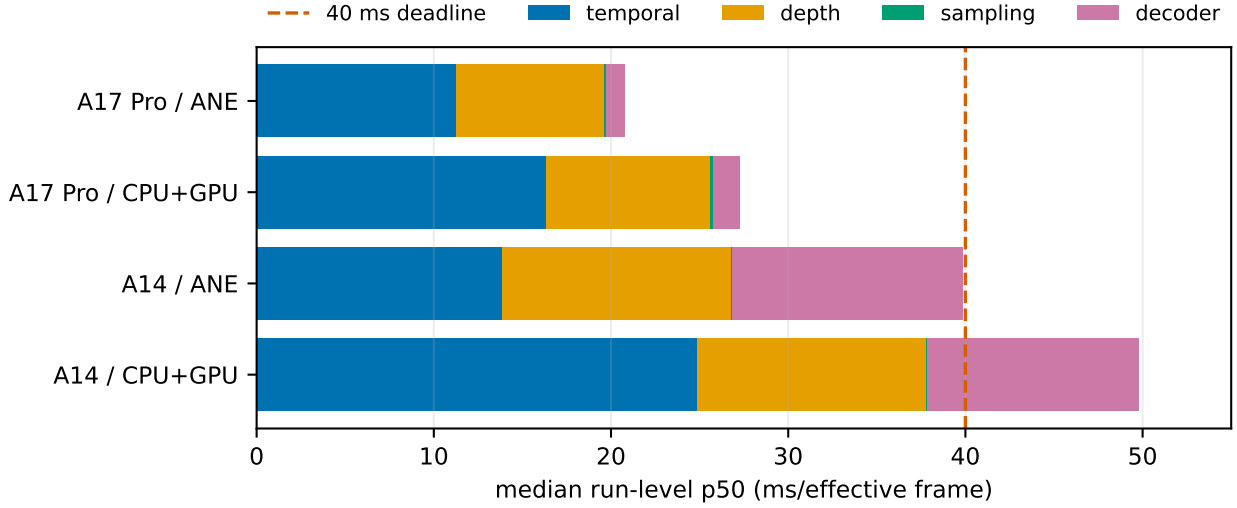


Figure 2: Stage decomposition for the repeated-process campaign. Bars use the median of run-level stage p50 values. The selected ANE policy reduces temporal cost on both phones; A14 reaches the deadline before its tail is considered. CPU+GPU denotes a requested Core ML policy, not a GPU-placement proof.

Five independent application processes per device and policy separate the selected ANE placement from a temporal .cpuAndGPU policy control; the prompt, seed, model hashes, depth CPU policy, decoder ANE policy, warmup count, and buffer settings are otherwise fixed. table 2 reports the median and IQR of each run-level percentile rather than pooling frames across processes.

On A17 Pro, ANE lowers median run-level p50 from 27.32 to 20.77 ms and p99 from 38.51 to 22.35 ms. The temporal stage alone falls from 16.35 to 11.25 ms p50; cold process startup to audible PCM falls from 6.22 to 4.26 seconds. On A14, ANE lowers p50 from 49.73 to 40.38 ms and the temporal stage from 24.85 to 13.85 ms (fig. 2). Its 50.72 ms p99 remains over deadline and is statistically indistinguishable from the GPU control’s 50.64 ms at this sample size. Thus the control supports an A17 tail-latency win and an A14 central-latency win; it does not turn A14 into a real-time device.

5.3 A17 Pro sustain: compute and delivery pass

The sustained device outcomes are compared in table 1. The final A17 run starts with the phone already in the **serious** thermal state after preceding experiments. It measures 610.186 seconds after generation starts and pulls 610.208 seconds of audio. The producer completes 629 one-second iterations, or 1.0308 times real time. Effective frame time is 20.260 ms p50, 20.813 ms p90, and 21.660 ms p99. There are no underruns and no dropped frames.

The finite reservoir does not create this pass (fig. 3). It begins at 138,240 frames (2.88 s), ends at 1,030,656 frames (21.47 s), and has a positive slope of 1,462.5 frames/s. After the initial fill, the per-minute p99 stays in a narrow

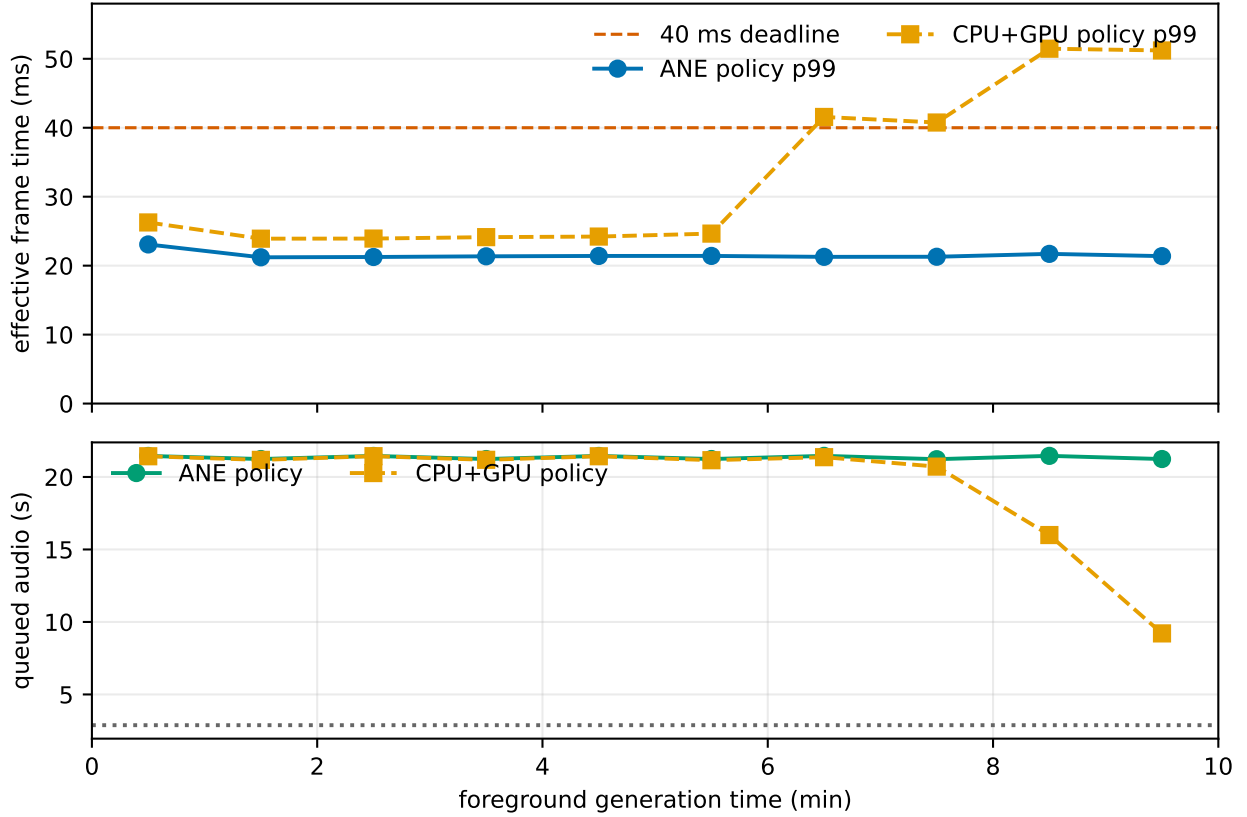


Figure 3: A17 Pro matched ten-minute trajectories. The selected ANE-policy run begins in **serious** thermal state and keeps p99 below deadline while queued audio grows. The CPU+GPU-policy control crosses the deadline after minute six and drains its banked reservoir, despite ending without an underrun. Both curves are generated from public receipts.

21.21-23.08 ms range. The first minute includes reservoir accumulation and has the largest p99; later minutes remain below 21.71 ms despite the sustained hot state. The compute clock and delivery clock therefore pass independently.

The matched 610.03-second `.cpuAndGPU` temporal-policy control separates that stability from mere buffered playback. It begins fair, reaches serious thermal state after 40.12 seconds, and ends with zero underruns, but p99 is 49.23 ms and production is only 1.008x. Its per-minute p99 crosses 40 ms after minute six and reaches 51.46 ms; a reservoir banked near 21 seconds drains to 7.65 seconds by the end. Thus its delivery clock happens to survive the finite experiment, while its compute clock fails. The selected ANE run begins hotter yet keeps tail latency flat, a stronger explanation than final underrun count alone.

5.4 A14 boundary: buffering cannot manufacture real time

The FP16 decoder configuration that works on A17 Pro is numerically unsafe on A14. ANE placement first yields finite but unusable amplitude; a long CPU retry later becomes logically non-finite. The final A14 experiment therefore uses the certified 136 MB FLOAT32 decoder with `.cpuAndNeuralEngine`, the same temporal ANE request, and CPU depth.

This configuration begins nominal, reaches **fair** at 51.08 s and **serious** at 106.06 s. Its effective frame time is 43.876 ms p50, 47.248 ms p90, and 58.586 ms p99. Production is 0.8967 times real time. The audio ring permits a maximum observed start reservoir of 20.16 seconds; it underflows at 294.08 seconds, records 7,952 callback underruns, and produces 568.32 seconds of PCM during the 610.04-second window. There are zero producer drops.

This is not an A14 real-time tier. It is a measured bounded-reservoir failure. The distinction matters: a short benchmark or a larger but unbounded preload could make the same system appear successful.

Table 3: Long-horizon A17 audio gate. Arrows show the required direction; bold values fail the frozen threshold. Lower temperature fixes only the pulse detector and is therefore rejected.

Arm	Clip ↓	L/R ↑	Prompt ↑	Ref. ↑	Pulse ↓	Blind
Threshold	10^{-5}	.970	.300	.800	.070	$\geq 4/5$
$T = 1.0$, 600 s	4.05×10^{-6}	.986	.312	.854	.136	2/5
$T = 0.5$, 600 s	3.27×10^{-5}	.946	.268	.784	.042	—

5.5 Generative clock: the long capture fails

The corrected temperature-1.0 A17 capture is 600.0 seconds, finite, 48 kHz stereo, and free of render failure. It passes clipping ($4.05\text{e-}6$), normalized chunk-boundary jump (0.0061), left/right correlation (0.9863), prompt adherence (0.3119), and reference embedding similarity (0.8541). All six known-bad controls are rejected.

It nevertheless fails the predeclared generative-quality gate in two independent ways. The last 30 seconds have 4-16 Hz envelope-pulse share 0.1364, nearly twice the 0.070 limit. All five blind votes are valid because every vote passes the clean reference and rejects the click control, but the candidate passes only two. The rejected excerpts begin at 151.68, 555.04, and 566.83 seconds; the calibrated judgments flag persistent periodic clicks. Accepted excerpts begin at 33.44 and 361.96 seconds. The defect is therefore intermittent across the trajectory, not a global recorder or render failure.

Two ablations prevent a facile explanation. A 10-second temporal-state refresh passes all bands at 180 seconds but not at 600 seconds. Lowering temperature to 0.5 reduces pulse share to 0.0417, but clipping rises to $3.27\text{e-}5$, left/right correlation falls to 0.9462, prompt adherence to 0.2684, and reference similarity to 0.7838. We reject the lower-temperature path rather than select the one metric it improves.

The two principal quality arms are compared in table 3. This is the principal negative result. ANE placement, p99 headroom, a growing ring, and semantically plausible embeddings do not guarantee sustained musical output. The system has solved inference and delivery, not long-horizon generation quality.

5.6 Compression ladder: smaller is not equivalent

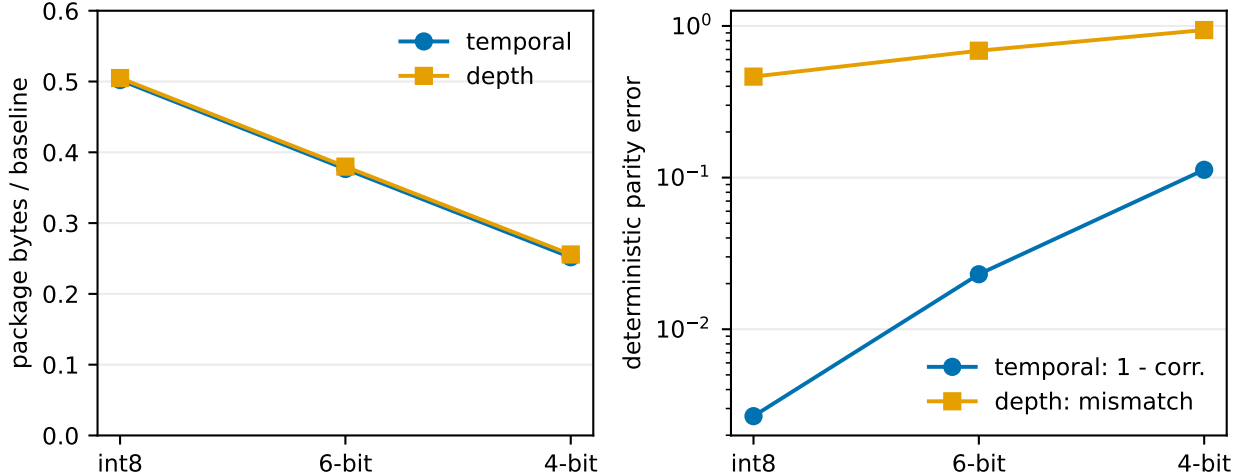


Figure 4: Rejected post-training compression ladder. Every artifact remains finite and becomes substantially smaller, but deterministic parity degrades monotonically. The gate order stops all six candidates before device timing or listening, avoiding a speed result for the wrong function.

The complete compression ladder appears in fig. 4. We test int8 linear quantization and 6- and 4-bit palettization separately on the 365.7 MB temporal and 74.5 MB depth packages. Temporal candidates shrink to 50.2%, 37.6%, and 25.2% of baseline. Their 64-step correlations are 0.9973, 0.9770, and 0.8874, below the 0.999 gate, with maximum errors 4.59, 13.21, and 23.95. Depth candidates shrink to 50.5%, 38.0%, and 25.6%; deterministic argmax mismatch

rises to 46.3%, 68.7%, and 94.0%.

All candidates remain finite. None reaches a phone. We therefore make no claim that compression improves latency or preserves audio, and the sustained system uses uncompressed temporal plus the prior FP16 depth artifact.

6 Discussion

6.1 The benchmark must follow the failure surface

The three clocks correspond to three different classes of bug. Compute failure is exposed by stage p99 and thermal trend. Delivery failure is exposed by ring depth, pushed/pulled accounting, and callback counters. Generative failure is exposed by full-horizon audio and calibrated controls. Substituting one class's instrument for another is invalid: an audio embedding cannot prove no underruns, and a zero-underrun log cannot prove music.

The stride bug demonstrates a fourth requirement: the evidence path must be tested too. Live playback already read the decoder tensor by declared strides; the recorder assumed dense storage and manufactured huge STFT coefficients from padding. Replaying the historically clean rolling model through the same recorder reproduced the corruption, falsifying the new temporal graph as the sole cause. One padded-FP16 unit test then closed the recorder defect. Without that falsification sequence, we might have "fixed" a correct renderer.

6.2 Accelerator placement is a runtime property

A convertible graph is not necessarily an admitted graph, and a requested policy is not observed execution. Our state boundary succeeded because it removed mutation from the compiled graph, but the contribution is not merely that this graph can compile. The 10-process matrix establishes repeatability in the application, while the model-attributed trace establishes physical ANE activity and GPU absence. Either artifact alone is weaker.

The broader design lesson is conservative: keep the compiled graph pure and fixed; keep small mutation and irregular lookup on the host; measure the whole pipeline. This resembles NPU partitioning in mobile LLM systems, but here the partition is constrained by a continuous audio deadline and by the need to prove state across predictions.

6.3 Negative results change the system boundary

The A14 result rules out a tempting product claim. A 20-second reservoir is large enough to make short demonstrations smooth, yet the full run exposes the sustained 0.897x rate. The compression ladder rules out a tempting engineering response: package bytes can be cut in half, but deterministic behavior changes before we have earned a speed measurement. The lower-temperature audio arm rules out another: suppressing one detector while damaging four other measures is not a fix.

Each failure narrows the next legitimate experiment. A14 needs a materially different precision/training or depth/decoder architecture, not a larger hidden buffer. The long-horizon quality defect needs token-level comparison against the reference sampler across the onset, not more audio-thread tuning. We stop where the evidence stops.

7 Limitations

This study uses one prompt, one principal sampling seed, two phones, and one model size. It characterizes a system and a failure mechanism, not the prompt distribution of MRT2. The A17 soak deliberately begins hot, which strengthens the sustain observation but is not an estimate of ordinary battery behavior. Instruments power-impact scores are not calibrated joules; we do not make a battery-life claim. A corrected, counterbalanced Power Profiler pair could not be completed because Instruments lost USB attachment to both phones after the signed bundle was repaired. The invalid preflight traces are excluded, and we report neither an energy comparison nor a process-impact comparison. The blind audio votes come from a multimodal model rather than human MUSHRA listeners, so they are treated only as a calibrated detector with explicit controls. The complete product runtime is private; public artifacts include exporters, validators, fixtures, pseudocode-level contracts, model hashes, normalized result manifests, the gate verifier, and this paper. The runtime is available from the author for artifact review.

Most importantly, the generative-quality gate fails. The title's "GPU-free MRT2" refers to observed compute placement and sustained inference, not to a claim that the current output is ready to ship for arbitrary-length

listening. A follow-up should reproduce the click onset with token taps, compare the Core ML and MLX sampled trajectories under identical noise, and localize the first divergence before changing another system parameter.

8 Conclusion

We demonstrate that the full compute and delivery path of a 230M-parameter live-music model can run GPU-free on an iPhone 15 Pro Max for ten hot foreground minutes with 1.031x production, 21.66 ms p99 effective frame cost, zero underruns, and a growing audio reservoir. A pure one-frame temporal graph plus host-owned K/V state makes ANE admission reliable without fragmenting the model. The same experiment draws a hard A14 boundary and rejects six attractive but invalid compressed artifacts.

It also demonstrates why those results are not the finish line. The 600-second output fails a frozen pulse detector and three of five calibrated blind votes even though performance, delivery, semantic, and stereo metrics pass. Live generation therefore has three clocks, not one. The scientifically correct result is both halves at once: sustained GPU-free inference on a phone is solved here; sustained generative quality is not.

9 Artifact statement

The public repository contains the Core ML exporters, frozen fixtures, validation scripts, machine-readable G1-G4 manifests and verifier reports, figure sources, manuscript source, model-weight hashes, and exact protocol. Preconverted public artifacts are mirrored on Hugging Face. Private raw device logs, Instruments traces, 600-second WAVs, and the Crossfade application are hashed by the public manifests and are available from the author for review.

10 Acknowledgments

OpenAI Codex assisted with experimental code, data normalization, figure generation, and manuscript preparation. The author designed the study, operated the devices, inspected the evidence, selected the claims, and accepts responsibility for the manuscript.

A Reproducibility

A.1 Public evidence map

The repository’s `validation/results/system-paper` tree is the canonical publication dataset. The `A17` and `A14 placement` directories contain G1 manifests and verifier reports; `a17pro/soak` contains G2; `audio` contains G3; `a14/soak` contains G4; and `evaluation/evaluation-manifest.json` contains the four five-process latency cells. `MRT2WeightCompressionLadder.json` contains the six early-stopped precision arms. Each manifest hashes the private raw event trace, console log, WAV, or Instruments export from which it was normalized. The verifier reports hash their input manifests, so a changed number invalidates the chain rather than silently updating a verdict.

The artifact identities that bind the principal runs are the same across the placement and sustain receipts: temporal weight SHA-256 prefix `02657e24`, depth `6c69ebae`, A17 FP16 decoder `ceb3ed7d`, and A14 FLOAT32 decoder `38cbdf5c`. Full hashes are retained in the manifests and model metadata; the prefixes are printed here only for readability. The 64-step temporal fixture contains 23 predictions after the 41-slot ring wraps. The public G1 report hash is itself included in G2, and the G2 report hash is included in G3.

A.2 Runtime protocol

The producer implements the following bounded protocol:

1. Load the signed, precompiled Core ML packages with explicit compute policies; query the temporal compute plan; run the fresh/warmed and 64-step state proofs; then execute three untimed warmups.
2. Prime a fixed-capacity single-producer/single-consumer PCM ring. For each temporal frame, present chronological K/V inputs, predict one temporal update, copy the 48 updates into the current ring slot, sample all 12 depth levels in one prediction, and sum the selected RVQ rows.

3. At decoder cadence, predict the STFT block, materialize it by declared shape and strides, perform inverse STFT and overlap-add, and push PCM. The audio callback performs only a bounded ring read; underflow is zero-filled and counted.
4. Every ten seconds, clear temporal K/V and previous temporal feedback while leaving decoded PCM, decoder overlap, and ring state continuous. Stop only after the measured horizon; capture events and rendered PCM independently.

The fixed condition is `warm ambient texture`, `warm.bin`, top-k 40, style guidance 3.1, temperature 1.0, seed 20260718, 48 kHz stereo, foreground, screen on. The principal A17 run begins already in the `serious` thermal state. The A14 run begins nominal and is allowed to transition naturally. The matched sustain control uses the same condition and differs only in requested temporal compute policy. A power pair was attempted under that design but produced no valid measurement, as stated in the limitations.

A.3 Re-running the public verdicts

The four hard verdicts require only Python and the checked-in manifests. Run the gate verifier with `g1` on the A17 placement manifest, then repeat with `g2`, `g3`, and `g4` on the corresponding sustain, audio, and A14 manifests. The relevant unit suites are:

- `test_system_paper_gates`;
- `test_system_paper_soak_manifest`;
- `test_system_paper_audio_manifest`; and
- `test_system_paper_evaluation_manifest`.

They exercise both accepted receipts and binding false positives. Figure sources read only public JSON. The manuscript builder regenerates LaTeX from this readable draft; Tectonic builds the PDF without a venue-specific class.

Device recapture additionally requires the private Crossfade host, the signed model bundle, and physical phones. The public exporter and fixture tests can still reproduce model boundaries, deterministic parity, compression rejection, manifest normalization, and every publication verdict without that host. This division is deliberate and is the limitation stated in the main text, not an implied open-source claim about the product runtime.

References

- Apple. Core ml tools. Software, 2026. URL <https://github.com/apple/coremltools>.
- Apple Machine Learning Research. Deploying transformers on the apple neural engine, 2022. URL <https://machinelearning.apple.com/research/neural-engine-transformers>.
- Antoine Caillon and Philippe Esling. RAVE: A variational autoencoder for fast and high-quality neural audio synthesis. *arXiv preprint arXiv:2111.05011*, 2021. URL <https://arxiv.org/abs/2111.05011>.
- Jade Copet, Felix Kreuk, Itai Gat, Tal Remez, David Kant, Gabriel Synnaeve, Yossi Adi, and Alexandre Défossez. Simple and controllable music generation. In *Advances in Neural Information Processing Systems*, volume 36, 2023. URL <https://arxiv.org/abs/2306.05284>.
- Alexandre Défossez, Jade Copet, Gabriel Synnaeve, and Yossi Adi. High fidelity neural audio compression. *Transactions on Machine Learning Research*, 2023. URL <https://arxiv.org/abs/2210.13438>.
- Google. Magenta realtime 2 model card, 2026. URL <https://huggingface.co/google/magenta-realtime-2>.
- Google Magenta. Magenta realtime 2. Open-weights model and inference system, 2026. URL <https://magenta.github.io/magenta-realtime/>.
- Awni Hannun, Jagrit Digani, Angelos Katharopoulos, and Ronan Collobert. MLX: Efficient and flexible machine learning on apple silicon. Software, 2023. URL <https://github.com/ml-explore/mlx>.
- Yunpeng Li, Kehang Han, Brian McWilliams, Zalán Borsos, and Marco Tagliasacchi. Spectrostream: A versatile neural codec for general audio. *arXiv preprint arXiv:2508.05207*, 2025. URL <https://arxiv.org/abs/2508.05207>.

- Lyria Team, Antoine Caillon, Brian McWilliams, Cassie Tarakajian, Ian Simon, Ilaria Manco, Jesse Engel, Noah Constant, Yunpeng Li, Timo I. Denk, et al. Live music models. *arXiv preprint arXiv:2508.04651*, 2025. URL <https://arxiv.org/abs/2508.04651>.
- Matt Mireles. Surgical inference: Decomposing generative models for heterogeneous on-device SoC inference. Companion technical manuscript and public artifact, 2026. URL <https://github.com/mattmireles/kokoro-coreml>.
- Vijay Janapa Reddi, David Kanter, Peter Mattson, Jared Duke, Thai Nguyen, Ramesh Chukka, Ken Shir-ing, Koan-Sin Tan, Mark Charlebois, William Chou, et al. MLPerf Mobile Inference Benchmark: An industry-standard open-source machine learning benchmark for on-device AI. In *Proceedings of Machine Learning and Systems*, volume 4, 2022. URL https://proceedings.mlsys.org/paper_files/paper/2022/hash/a2b2702ea7e682c5ea2c20e8f71efb0c-Abstract.html.
- Daliang Xu, Hao Zhang, Liming Yang, Ruiqi Liu, Gang Huang, Mengwei Xu, and Xuanzhe Liu. Fast on-device LLM inference with NPUs. In *Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems*, 2025. doi: 10.1145/3669940.3707239. URL <https://arxiv.org/abs/2407.05858>.
- Neil Zeghidour, Alejandro Luebs, Ahmed Omran, Jan Skoglund, and Marco Tagliasacchi. Soundstream: An end-to-end neural audio codec. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:495–507, 2022. doi: 10.1109/TASLP.2021.3129994. URL <https://arxiv.org/abs/2107.03312>.